

Lecture 9 - DS

15 May 2026
Zerand

Planar dynamical systems.

$\dot{x} = f(x)$ general notation, $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f \rightarrow C^1$ function.

$m=1 \Rightarrow \dot{x} = f(t, x)$

$m=2 \Rightarrow \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = A \cdot \begin{pmatrix} x \\ y \end{pmatrix}$

Not all DE can be solved, but in those cases we use $\exists!$ theorem.

$\varphi(t, \eta) = (x(t, \eta), y(t, \eta))$: $m=2 \leftarrow$ the FLOW for a system

(the solution is a pair) $\rightarrow$ a curve.

$\mathcal{S}_\eta = \{ \varphi(t, \eta) : t \in \mathbb{I}_\eta \}$

$m=2$ $\dot{x} = Ax$ (general form of a linear system)

$\det A \neq 0$

$\det A = \lambda_1 \cdot \lambda_2$

$\det(A - \lambda I_2) = 0$. if A is not diagonal

λ_1, λ_2 eigenvalues if A is diagonal.

NODE $0 < \lambda_1 \leq \lambda_2$ or $\lambda_1 \leq \lambda_2 < 0$

SADDLE $\lambda_1 < 0 < \lambda_2$

CENTER $\lambda_{1,2} = \pm i\beta$

FOCUS $\lambda_{1,2} = \alpha \pm i\beta$

- system classification (λ)
- quadrants $\rightarrow$ value $\rightarrow$ arrow
- $\text{Re}(\lambda) > 0 \Rightarrow$ trigon. $\hookrightarrow$
- $< 0 \Rightarrow$ dec. $\hookrightarrow$
- global attr ($\lim_{t \rightarrow \infty} \varphi(t, \eta) = \text{attr}$)
- F.I.
- $\theta < 0 \Rightarrow$ clock.
- $\theta > 0 \Rightarrow$ trig.

• take $\mathbb{I}$ quadrants $\rightarrow$ take for ex. $x=2, y=2 \} \Rightarrow$ arrow to the right

How to draw phase portrait in 2D:

take each quadrant

$\downarrow$
values in each one

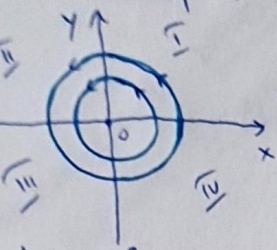
$\begin{matrix} x < 0, & y > 0 \\ x > 0, & y > 0 \\ \text{(dec.)} & \text{(incr.)} \end{matrix}$

$\rightarrow$ take $x=-2, y=2$

$\begin{matrix} x < 0, & y < 0 \\ x > 0, & y < 0 \end{matrix}$

$\} \Rightarrow$ arrow ~~to~~ down.

CENTER ($\lambda_{1,2} = \pm i$)
(circles/ellipses)

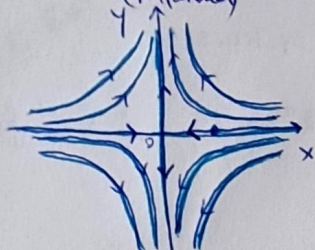


$$\lambda_{1,2} = \pm i$$

$$\dot{x} = -y$$

$$\dot{y} = x$$

SADDLE ($\lambda_1 = -1, \lambda_2 = 1$)
(hyperbolas)



I:

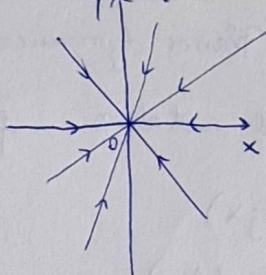
on the axis: $y=0, x=2$

$$\dot{x} = -2, \dot{y} = 0$$

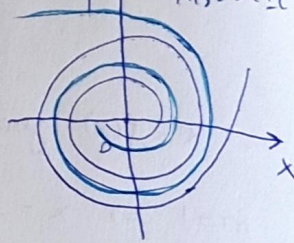
$x \downarrow$

arrow $\leftarrow$

NODE ($\lambda_1 = -1, \lambda_2 = -1$)
(rays)



FOCUS (spirals)
 $\lambda_{1,2} = 1 \pm i$



(left) trigonou.
real part $> 0 \Rightarrow$ (outward spiral)
real part $< 0 \Rightarrow$ (inward spiral)
(right) clockwise

c) find flow. def. attractor, origin = global attractor.

$$\phi(t, \eta_1, \eta_2) = (\eta_1 \cdot e^{-t}, \eta_2 \cdot e^{-t})$$

$$\lim_{t \rightarrow \infty} \phi(t, \eta_1, \eta_2) = (0, 0)$$

$$\dot{x} = -x \quad \dot{y} = -y \quad x(0) = \eta_1 \quad y(0) = \eta_2$$

$$\frac{dx}{dt} = -x \quad \frac{dy}{dt} = -y$$

$$\frac{dx}{x} = -dt \quad \int$$

$$\int \frac{1}{x} dx = \int -dt$$

$$\ln|x| = -t + C$$

$$x(t) = e^{-t} \cdot C$$

$$x(0) = C = \eta_1$$

$$x(t, \eta_1) = \eta_1 \cdot e^{-t} \quad y(t, \eta_2) = \eta_2 \cdot e^{-t}$$

Prove $\mathbb{R} \times (0, \infty)$ is an invariant set.

$$\eta_1 \times \eta_2 \in \mathbb{R} \times (0, \infty)$$

$$\Rightarrow \eta_2 > 0 \Rightarrow \eta_2 \cdot e^{-t} > 0$$

$(\eta_1, \eta_2) \rightarrow$ all the orbits are included there.
starting points

$\eta_1 \rightarrow$ anywhere on Ox (I, II quadrant)

1st integral (always equal to a constant)

in $\mathbb{R} \times (0, \infty)$:

$H: \mathbb{R}^2 \rightarrow \mathbb{R}$, $H(x, y)$ does not depend on time.

$H(x, y) = \frac{y}{x}$ or $\frac{x}{y}$ (in our example)

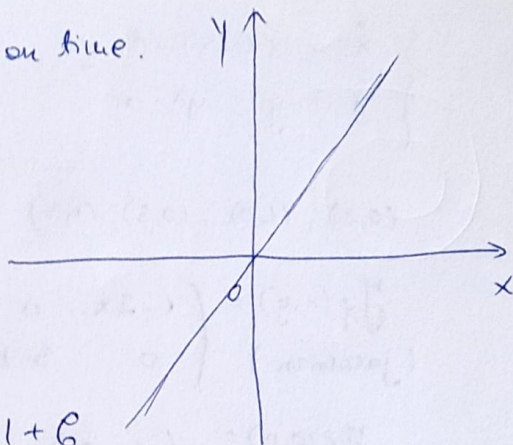
$$\boxed{\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}} = \frac{-y}{-x} = \frac{y}{x}}$$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{x} \quad | \int$$

$$\Rightarrow \int \frac{1}{y} dy = \int \frac{1}{x} dx \Rightarrow \ln|y| = \ln|x| + C$$

$$\Rightarrow \ln\left|\frac{y}{x}\right| = C \Rightarrow \frac{y}{x} = C = H(x, y)$$

$H(x, y) = \frac{y}{x} \Rightarrow y = \frac{1}{c}x$ line which passes the origin
(good to sketch the graph)



HW: First integral for each graph $\rightarrow H(x, y) = c \Rightarrow y$ in terms of $x \Rightarrow$
 $\Rightarrow$ what kind of function $\Rightarrow$ draw the phase portrait.

• Focus

$\theta < 0 \Rightarrow$ right.
anti-trig.

$\theta > 0 \Rightarrow$ left.
trig.

Semi stable $\rightarrow$ one part attractor
 $\rightarrow$ one part repeller.

A diagonalizable: Jordan matrix = diag. vector.

$\exists P$ s.t. $A = PDP^{-1} \Rightarrow A$ diagonalizable

over $\mathbb{C}^n \rightarrow$ can have complex numbers.

P should be invertible.

u
v
w
m

lin. method. on non-linear SYS. $\rightarrow$ stability?

$$\begin{cases} \dot{x} = x(1-x) = 0 \\ \dot{y} = y(3-y) = 0 \end{cases}$$

$$(0,0), (1,0), (0,3), (1,3)$$

$$J_f(x,y) = \begin{pmatrix} 1-2x & 0 \\ 0 & 3-2y \end{pmatrix}$$

(jacobian)

$$J_f(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \Rightarrow \lambda_1 = 1 \in \mathbb{R}, > 0 \quad \lambda_2 = 3 \in \mathbb{R}, > 0 \quad \Rightarrow (0,0) \text{ unstable, global repeller.}$$

NODE.

$$J_f(1,0) = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \Rightarrow (1,0) \text{ unstable SADDLE.}$$

! second-order d.e.

$\hookrightarrow$ system of two first-order d.e.

$$\begin{aligned} \cancel{y}, \quad \cancel{\dot{y} = x} \quad \dot{x} &= y \\ y \quad \dot{y} &= \ddot{x} = f(x, \dot{x}) = f(x, y). \end{aligned}$$

$$\begin{cases} x = \theta \\ y = \dot{\theta} \end{cases}$$

$$\ddot{\theta} + \omega^2 \sin \theta = 0$$

$$x = \theta$$

$$y = \dot{x}$$

$$\dot{x} = y$$

$$\dot{y} = \ddot{x} = \ddot{\theta} = -\omega^2 \sin \theta$$